

Lance Main

Data Analyst

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PROFESSIONAL SUMMARY

Data Analyst with 4+ years of experience turning data into actionable insights and strategic solutions. Proficient in Python, R, SQL, and experienced with IDEs like Visual Studio Code. Skilled in data visualization with Tableau, Power BI, and Excel. Adept at managing databases (MySQL, PostgreSQL, SQL Server, MongoDB) and optimizing data integrity. Experienced in leveraging AWS services (S3, RDS, Glue) to enhance data processing. Strong in data cleaning, critical thinking, and cross-functional collaboration, with a proven ability to drive impactful customer segmentation and strategic initiatives. Skilled in Git and GitHub, adaptable to Windows and Linux.

SKILLS & CERTIFICATIONS

Methodologies :	SDLC(Agile, Scrum, Waterfall, Kanban)
Programming Language :	Python, R, SQL, VBA, Shell Scripting
IDE's :	Jupyter Notebook, R Studio, Visual Studio Code, PyCharm
Python Packages :	NumPy, Pandas, Matplotlib, SciPy, ggplot2, Scikit learn, PySpark, Keras, TensorFlow
Visualization & ETL Tools :	Google Analytics, Elastic, Tableau, Power BI, Talend, Airflow, SSIS, Alteryx, Informatica
Database :	PostgreSQL, Microsoft SQL Server, MySQL, Oracle RDBMS
Cloud Technologies :	Databricks, Azure, AWS (S3, RDS, Redshift, Lambda, Glue, Snowflake
Other Tools :	Docker, MS Excel, Google Sheets, Git, Looker, GitHub, MS Visio, Qlik, Jira, CI/CD, Git Bash
Operating System :	Windows, Linux
Certifications :	Google Data Analytics Certificate

WORK EXPERIENCE

Data Analyst | [American International Group, MA](#)

January 2024 – Present

- Developed a predictive churn model using XGBoost, improving accuracy by 20% and enabling targeted customer retention strategies.
- Applied machine learning frameworks including TensorFlow, Keras, and Scikit-learn to solve complex business problems across customer engagement and financial forecasting.
- Utilized NLP on customer communication logs and financial documentation to extract behavioral and transactional insights, enhancing segmentation strategies by 25%.
- Employed Decision Trees and Random Forests for predictive analytics, driving a 35% reduction in churn and a 20% drop in SLA violations through data-driven interventions.
- Automated end-to-end data pipelines using Python, SQL, Pandas, and NumPy, cutting financial data validation and reconciliation efforts by 50% and reducing overall data preparation time by 15%.
- Developed and deployed Tableau and Power BI dashboards to visualize shipment analytics, SLA breaches, KPI variances, and cost trends—enabling strategic decisions across finance and operations.
- Built and maintained data pipelines in VS Code and SSIS to streamline cross-departmental purchase order and contract validation, reducing discrepancies and enhancing compliance.
- Deployed machine learning models on AWS using Lambda, S3, and Glue, ensuring scalable, secure, and efficient inference workflows across multiple data platforms.
- Containerized data applications using Docker and orchestrated CI/CD pipelines via GitLab, decreasing deployment time by 40% and ensuring environment consistency.
- Integrated AWS RDS and CloudWatch to improve financial data reporting, real-time system monitoring, and early anomaly detection elevating cross-team transparency and accountability.

Data Analyst | [Spectrum Webapps, IND](#)

April 2020 – August 2022

- Developed a real-time anomaly detection and predictive maintenance system using PyTorch, Scikit-Learn, Prophet, and SciPy, reducing equipment downtime by 25% and improving operational efficiency by 30%.
- Enhanced financial anomaly detection by 30% through the design and deployment of LSTM and Variational Autoencoder (VAE) models, enabling proactive fraud prevention in high-volume transactions.
- Improved time-series forecasting accuracy by 30% using RNN and LSTM architectures to analyze sequential financial data for cash flow and expense predictions.
- Conducted in-depth financial data analysis using Python, PySpark, and SQL, identifying discrepancies and implementing corrective actions that increased reporting accuracy and reduced audit risks.
- Built scalable ETL pipelines with Pandas, Spark, and GCP BigQuery, optimizing data preprocessing and model training workflows and reducing data processing time significantly.
- Developed interactive dashboards in Tableau and Power BI to monitor KPIs, vendor spend, contract compliance, and forecasting insights for leadership teams in finance and procurement.
- Improved invoice validation and reconciliation efficiency by 40% by automating workflows in PySpark and Python, saving over 15 hours of manual effort weekly.
- Supported procurement and finance teams in verifying purchase orders, contract terms, and payment schedules, reducing billing disputes and improving vendor relations.
- Led data quality improvement initiatives using advanced SQL and Google Sheets-based tools, boosting accuracy in monthly financial reporting by over 25%.

KEY PROJECTS

ASTRE BIKE SHARE DASHBOARD DEVELOPMENT (SQL Server, SSMS, Power BI, DAX, Excel)

- Architected and implemented a relational database in SQL Server, using SQL Server Management Studio (SSMS) for advanced data management. Data was ingested from structured Excel files into normalized tables, ensuring efficient storage and retrieval.
- Engineered complex SQL queries for data transformation, cleansing, and aggregation, utilizing advanced SQL techniques such as CTEs, window functions, and subqueries to enhance data quality and integrity.
- Integrated SQL Server with Power BI to create a dynamic and interactive dashboard that led to a 25% improvement in decision-making accuracy. Utilized Data Analysis Expressions (DAX) to build calculated columns and measures for in-depth analysis and visualization.
- Developed key performance indicators (KPIs) to monitor hourly revenue, seasonal trends, and rider demographics, which improved operational planning and resource allocation by 20%.
- Designed and customized the dashboard using Power BI's advanced formatting options, incorporating corporate branding elements and ensuring a user-centric interface for seamless navigation.

EMPLOYEE CHURN PREDICTION USING GOOGLE CLOUD PLATFORM (GCP, BigQuery, Python, PyCaret, Looker Studio)

- Developed a cloud-based data pipeline on Google Cloud Platform (GCP) using BigQuery for data storage and management of employee-related metrics.
- Built a predictive model with PyCaret to identify employees likely to churn, utilizing a 70/30 train-test split. Key predictors like satisfaction level and salary were identified through feature engineering.
- Deployed the model on GCP for real-time predictions, integrated with cloud services for ongoing monitoring.
- Created an interactive dashboard in Looker Studio, showcasing KPIs related to churn with visualizations such as scorecards and bar charts.
- Recommended strategies to reduce churn, including employee recognition, training programs, and salary adjustments.

DATA ANALYSIS OF COVID-19 IMPACT IN INDIA (Tech Stack: MS Excel, SQL, Python, R, PowerBI)

- Collected and cleaned large datasets from various sources, including government websites, news articles, and social media platforms, for analysis of COVID-19 cases, deaths, and vaccination rates.
- Developed interactive visualizations by implementing software such as PowerBI and Python's Matplotlib to present trends in COVID-19 cases, deaths, and vaccination rates at national, state, and county levels.

EDUCATION

Master of Science in Information Systems | Northeastern University, Boston, MA

September 2022 – August 2024

(Relevant Courses: Business Analysis & Information Engineering, Database Management & Design, Planning Information Systems)

Bachelor of Engineering in Information Technology | University of Mumbai

August 2018 – May 2022

(Relevant Courses: Big Data Analytics, Data Structures, Project Management, Data Mining & Business Intelligence, Cloud Computing)